

Arnav Surve

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EDUCATION

California State University - Sacramento

Bachelor of Science, Computer Science

Sacramento, CA

Aug. 2024 – May 2026

Purdue University

Bachelor of Science, Cybersecurity – Dean's List, Fall 2023

West Lafayette, IN

Aug. 2022 – May 2024

EXPERIENCE

Software Engineer Intern

California Department of Motor Vehicles

March 2025 – Present

Sacramento, CA

- Developed greenfield Java Spring internal service to modernize mainframe-based vehicle registration sticker printing, replacing outdated technician interfaces and streamlining document generation pipelines, processing **\$10 billion** in annual revenue and printing **3 million** registration stickers monthly.
- Architected and deployed end-to-end ETL pipeline using AWS Lambda, Glue, and Apache Spark, processing mainframe VSAM dataset exports for vehicle registrations and driver licenses, successfully migrating **198 million** mainframe records to production RDS PostgreSQL database and handling **200-300k** daily records with upsert/delete operations serving **California Highway Patrol, FBI**, and other federal law enforcement agencies.
- Implemented entire infrastructure as code using Terraform, managing deployment lifecycle across dev, UAT, and production environments while supporting comprehensive UAT testing throughout development cycle.
- Refactored and deployed distributed vehicle title management microservices blockchain system using TypeScript monorepo with 5 Tezos smart contracts (FA2 NFTs), implementing multitenant role-based access control and chain bridge worker pools for database-blockchain synchronization. Supporting **45 million** vehicle title NFTs in production.
- Deployed self-hosted private Tezos blockchain node infrastructure via Kubernetes using ECR and EKS, orchestrating containerized microservices with ArgoCD pipelines and tiered monitoring.

IT Infrastructure Intern

MITER Brands – Milgard Manufacturing

June 2024 – Aug. 2024

Tacoma, WA

- Implemented APM pipeline optimizations in Site24x7 and New Relic, reducing alert redundancy by **60%** across 6 teams and 2 business units.
- Developed REST API endpoint monitors identifying critical vendor discrepancies, resulting in an average **20%** latency reduction across hot paths.
- Automated Active Directory compliance audits using PowerShell scripting, reducing manual audit time by **75%**.

PROJECTS

Grace Programming Language | *Go, Compiler Design, Type Systems, Code Generation*

2024

- Built statically-typed programming language (8,407 LOC) that compiles to readable COBOL, featuring complete compiler toolchain with lexer, recursive descent parser using Pratt parsing, and comprehensive AST for semantic analysis.
- Designed novel width-aware type system with compile-time COBOL PIC clause validation, automatic width inference for arithmetic expressions, and overflow detection for mainframe compatibility.
- Implemented code generation translating modern constructs (functions, structs, type inference) into structured COBOL with automatic temporary variable management.

Grace Orchestration Engine (graceinfra.org) | *Go, Zowe CLI, DAG Execution, Mainframe Modernization*

2024

- Engineered hybrid mainframe-cloud orchestration framework (8,366 LOC) modernizing mainframe workflows through declarative YAML configuration and Infrastructure-as-Code principles.
- Built DAG execution engine with concurrent job processing, dependency resolution, dynamic file/dataset resolution, and intelligent state management for complex multi-step mainframe-cloud workflows.
- Implemented extensible plugin architecture supporting multiple job handlers (compile, linkedit, execute, shell) with automatic cleanup of temporary mainframe artifacts and comprehensive error handling.

SKILLS

Languages: Go, Python, TypeScript, JavaScript, SQL, PostgreSQL, Java, HTML/CSS, YAML, JSON, Bash

Frameworks: React, Node.js, FastAPI, Express.js, Echo, GORM, GraphQL, Prisma, Model Context Protocol

Cloud/DevOps: AWS, Lambda, Glue, RDS, ECR, EKS, Kubernetes, Docker, Terraform, Git, GitHub Actions

Data/Blockchain: PostgreSQL, Redis, ETL, VSAM, Blockchain, Smart Contracts, Tezos, NFTs, Web3

Practices: CI/CD, Microservices, REST APIs, Agile/Scrum, DevOps, Monitoring, IaC